Cryptography: Step-by-Step Numericals Guide

Professor's Tip: *Beta, paper mein calculation hamesha likh kar karna. Modulo (Mod) ka asaan matlab "Remainder" (Baqi bachne wala hissa) hota hai. A = (B × Q) + R, kahan R apka answer (Mod) hai. A to Z ki numbering 0 se 25 tak yaad rakhna (A=0, B=1, ... Z=25).*

1. Euclidean Algorithm (Finding GCD)

Sawal: GCD(252, 105) find karein. (Maksad: Sab se bara common divisor nikalna)

Formula: A = (B × Quotient) + Remainder Step 1: 252 ko 105 se divide karo. 252 = 105 × 2 + 42 (Kiyunke 105 x 2 = 210. 252 - 210 = 42 remainder) Ab nayi values: A ban gaya 105, aur B ban gaya 42. Step 2: 105 ko 42 se divide karo. 105 = 42 × 2 + 21 (Kiyunke 42 x 2 = 84. 105 - 84 = 21 remainder) Ab nayi values: A ban gaya 42, aur B ban gaya 21. Step 3: 42 ko 21 se divide karo. 42 = 21 × 2 + 0 (Remainder 0 aa gaya!) Result: Jis divisor par remainder 0 aa jaye, woh GCD hota hai. Answer = 21

2. Caesar Cipher & Vigenère Cipher Math

Caesar Cipher Sawal: Plaintext = "HELLO", Key = 3. Encrypt karein.

Formula: C = (P + K) mod 26 Values: H=7, E=4, L=11, L=11, O=14. Key(K) = 3. H: (7 + 3) mod 26 = 10 mod 26 = 10 -> K E: (4 + 3) mod 26 = 7 mod 26 = 7 -> H L: (11 + 3) mod 26 = 14 mod 26 = 14 -> O L: (11 + 3) mod 26 = 14 mod 26 = 14 -> O O: (14 + 3) mod 26 = 17 mod 26 = 17 -> R Answer: KHOOR

Vigenère Cipher Sawal: Plaintext = "HELLO", Keyword = "KEY".

Rule: Keyword ko plaintext ki length jitna lamba karo: K, E, Y, K, E Formula: $C_i = (P_i + K_i) \bmod 26$ Letter 1 (H+K): H(7) + K(10) = 17 mod 26 = 17 -> R Letter 2 (E+E): E(4) + E(4) = 8 mod 26 = 8 -> I Letter 3 (L+Y): L(11) + Y(24)= 35. Ab 35 mod 26 nikalo: 35 - 26 = 9 -> J Letter 4 (L+K): L(11) + K(10)= 21 mod 26 = 21 -> V Letter 5 (O+E): O(14) + E(4) = 18 mod 26 = 18 -> S Answer: RIJVS

3. Playfair Cipher (Matrix & Rules)

Sawal: Keyword = "MONARCHY". Plaintext = "HE LL O" ko encrypt karein. (I/J combined).

Step 1: Matrix banayein. Pehle keyword ke letters likhein (without repeat), phir baqi alphabets A-Z likhein.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

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Step 2: Plaintext ko 2-2 ke pairs mein torein. "HELLO" -> HE, LL, O Rule: Same letters ek pair mein nahi aa sakte. 'LL' ke beech 'X' lagao. Naye Pairs: HE, LX, LO Step 3: Rules apply karein matrix dekh kar: Pair 1: (H, E) -> H aur E same column mein hain (Column 2). Rule: Niche wala letter lo. H ke niche F, E ke niche P. -> "CF" (Wait, E ke niche L hai, H ke niche F. Dhyan se! H(row 2, col 2) -> neeche F. E(row 3, col 1) -> neeche L. Nahi, yeh rectangle rule banega!) Correction: H aur E na same row mein hain, na same col mein. Rectangle ban raha hai. H ke samne E wale column mein: C. E ke samne H wale column mein: F. -> Pair 1 = CF. Pair 2: (L, X) -> Rectangle. L ke col/X ki row: U. X ke col/L ki row: S. -> SU Answer for first two pairs: CF SU...
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4. DES: S-Box Calculation (The Heart of DES)

Sawal: S-Box mein 6-bit input "101100" ja raha hai. Output 4-bit kya hoga? (Assume S-Box 1 table hai).

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Rule: 6-bit input ko torenge 2 hisson mein. Input = 1 0 1 1 0 0 Step 1: Row (Kitar) Number Pehla bit (1) aur akhri bit (0) ko milao -> "10" Binary "10" decimal mein kya hota hai? 2. Iska matlab hum S-Box ki Row Number 2 dekhein ge (Rows 0, 1, 2, 3 hoti hain). Step 2: Column (Caalam) Number Bich wale 4 bits ko lo -> "0110" Binary "0110" decimal mein kya hota hai? (4 + 2 = 6). Iska matlab hum Column Number 6 dekhein ge. Step 3: Table Lookup S-Box 1 table mein Row 2 aur Column 6 par jo value rakhi hogi (maslan wahan '9' rakha hai), usay 4-bit binary mein likh denge. Decimal 9 = 1001. Answer = 1001 (6-bit input ab 4-bit ban chuka hai).
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5. AES: ShiftRows Operation Matrix

Sawal: Niche diye gaye 4x4 State matrix par AES ka "ShiftRows" step apply karein.

Input Matrix (Hex Bytes)			
A0	B0	C0	D0
A1	B1	C1	D1
A2	B2	C2	D2
A3	B3	C3	D3

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Output Matrix (After ShiftRows)			
A0	B0	C0	D0
B1	C1	D1	A1
C2	D2	A2	B2
D3	A3	B3	C3

Logic & Explanation (Har Row ka rule): Row 0: 0 bytes left shift. (Koi change nahi aayega). [A0, B0, C0, D0] waise ka waise hi raha. Row 1: 1 byte left shift. (Pehla element uth kar aakhir mein chala jayega). [A1, B1, C1, D1] ban jayega -> [B1, C1, D1, A1] Row 2: 2 bytes left shift. (Pehle do elements uth kar aakhir mein). [A2, B2, C2, D2] ban jayega -> [C2, D2, A2, B2] Row 3: 3 bytes left shift. (Pehle teen elements uth kar aakhir mein). [A3, B3, C3, D3] ban jayega -> [D3, A3, B3, C3]

6. CBC Mode (Cipher Block Chaining) XOR Math

Sawal: CBC Mode mein P1 (Block 1) = "1010", IV = "1100", aur Key = "0101" hai. C1 (Ciphertext 1) nikalain.

Formula for C1: C1 = Encrypt(Key, P1 ⊕ IV) Rule of XOR (⊕): Same bits hon to 0, Different bits hon to 1. Step 1: P1 aur IV ko XOR karo. P1 = 1 0 1 0 IV = 1 1 0 0 ------------- (XOR) R1 = 0 1 1 0 (Bolo: Different to 1, same to 0) Step 2: Ab is R1 ko Key ke sath Encrypt karo. (Note: Example asaan karne ke liye hum man lete hain ke "Encrypt" ka matlab yahan Key ke sath dubara XOR karna hai, jabke asal mein DES/AES use hota hai). R1 = 0 1 1 0 Key = 0 1 0 1 ------------- (XOR) C1 = 0 0 1 1 Answer: Pehle block ka Ciphertext (C1) = "0011". Agle block P2 ko encrypt karne ke liye ab is C1 ko IV ki jagah use kiya jayega.

7. RC4 Stream Cipher: KSA (Key-Scheduling) Simulation

Sawal: RC4 algorithm mein maan lo S array sirf 4 size ka hai (0 to 3) aur Key = [1, 5] hai. Pehli 2 iterations ke baad S array ki halat kya hogi?

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Initial Setup: S array (Size 4) = [0, 1, 2, 3] T array (Key repeat karni hai
4 pora karne tak) = [1, 5, 1, 5] Variable: j = 0 Formula: j = (j + S[i] + T[i]) mod
4 --- Iteration 1 (jab i = 0): --- j = (0 + S[0] + T[0]) mod 4 j = (0 + 0 + 1) mod
4 = 1 mod 4 = 1. Ab S[0] aur S[j] (yani S[1]) ko swap (apas mein tabdeel) kar do. S
tha: [0, 1, 2, 3]. S[0] ki jagah 1, aur S[1] ki jagah 0 aa gaya. Naya S = [1, 0, 2,
3] --- Iteration 2 (jab i = 1): --- Ab purane wala j=1 use hoga, aur Naya S array
dekhna hai. j = (1 + S[1] + T[1]) mod 4 j = (1 + 0 + 5) mod 4 = 6 mod 4 = 2. Ab
S[1] aur S[j] (yani S[2]) ko swap karo. S[1] par 0 hai, S[2] par 2 hai. In dono ko
tabdeel karo. Naya S = [1, 2, 0, 3] Answer: 2 iterations ke baad S array = [1, 2,
0, 3] hoga.
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📝 Ab Aap Ki Practice Ka Waqt Hai (Self-Test)

Beta, exam mein janey se pehle in sawalon ko khud hath se solve karke dekhna:

1. **GCD:** Euclidean method se GCD(300, 18) find karein. (Hint: Remainder zero kab aata hai?)
2. **Caesar:** "ATTACK" ko Key=5 se encrypt karein, aur wapas decrypt karein. (A=0, B=1)
3. **Vigenère:** Plaintext="CRYPTO", Keyword="CAT". Encypt karein (Formula: C = (P+K) mod 26).
4. **Playfair:** Keyword "STUDENT" se 5x5 matrix banayein, aur "EXAM" (EX, AM) ko encrypt karein. (Yaad rakhein I aur J ek dabbe mein hain).
5. **DES S-Box:** Agar S-Box ka 6-bit input "110111" ho, toh Row number aur Column number kya banega? (Hint: pehla/akhri mila kar Row banti hai).
6. **XOR Practice:** "11001100" ko "10101010" ke sath XOR karein. (Dono same hon to 0, mukhtalif hon to 1 likhein).

> *In practice sawal ko apne class fellows ke sath mil kar verify karna. Best of luck beta, tension nahi leni, math aasan hai!*